CONVOLUZIONE

Def: Un SOTTOINSIEME di $\mathbb{R}^N$, K si dice COMPATTO se:

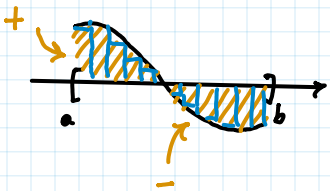
→ CHIUSO (complementare di un aperto)

→ LIMITATO: $\exists\, \delta > 0,\ \delta \in \mathbb{R}$ t.c. $\forall\, \underline{x}, \underline{y} \in K \quad \|\underline{x} - \underline{y}\| < \delta$

Se $[a,b] \subseteq \mathbb{R}$, $[a,b]$ è COMPATTO e si può definire

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

thm fondamentale calcolo integrale



Def: Si dice SCATOLA l'insieme $[\underline{a}, \underline{b}] \subseteq \mathbb{R}^N$:

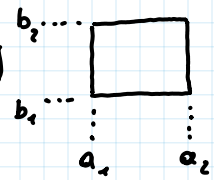
$$[\underline{a}, \underline{b}] = \{ \underline{x} \in \mathbb{R}^N : a_1 \le x_1 \le b_1 \wedge a_2 \le x_2 \le b_2 \wedge \dots \wedge a_N \le x_N \le b_N \}$$

$$= [a_1, b_1] \times [a_2, b_2] \times \dots \times [a_N, b_N]$$

Def: Si dice FUNZIONE CARATTERISTICA o INDICATRICE di un INSIEME $A \subseteq \mathbb{R}^N$:

$$\mathbb{1}_A(\underline{x}) = \begin{cases} 1 & \text{se } \underline{x} \in A \\ 0 & \text{altrimenti} \end{cases}$$

$N=2$ Scatola: $[a_1, b_1] \times [a_2, b_2] \subseteq \mathbb{R}^2$



$\mathbb{1}_{[a_1,b_1]\times[a_2,b_2]} \longrightarrow$

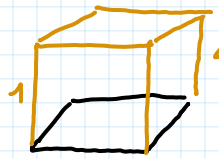
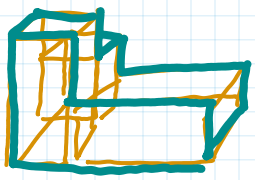


grafico della fz caratt.

Def: Si dice FUNZIONE SEMPLICE ELEMENTARE $\varphi : \mathbb{R}^N \to \mathbb{R}$

$$\varphi(\underline{x}) = \sum_{i=1}^{m} \lambda_i \cdot \mathbb{1}_{[\underline{a}_i, \underline{b}_i]}(\underline{x})$$

$\lambda_i \in \mathbb{R}$



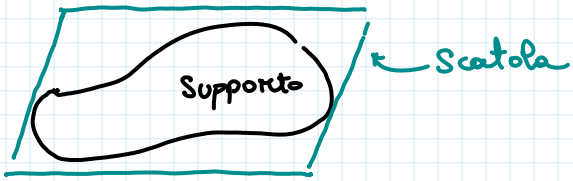
Si dice che $\varphi \in S$.

Sia $f : \mathbb{R}^N \to \mathbb{R}$ limitata e a SUPPORTO COMPATTO :

$$S_-(f) = \{ \varphi \in S : \varphi(\underline{x}) \leq f(\underline{x}) \quad \forall \underline{x} \in \mathbb{R}^N \}$$ MINORANTI di f

$$S_+(f) = \{ \varphi \in S : \varphi(\underline{x}) \geq f(\underline{x}) \quad \forall \underline{x} \in \mathbb{R}^N \}$$ MAGGIORANTI di f

OSS: $S_-(f) \neq \emptyset$ e $S_+(f) \neq \emptyset$ perchè f è LIMITATA e SUPP. COMPATTO



f è limitata: $|f(\underline{x})| < \delta$

Def: Sia $\varphi \in S$ allora si dice INTEGRALE di φ su $\mathbb{R}^N$ il NUMERO:

$$\int_{\mathbb{R}^N} \varphi(\underline{x}) \boxed{d\underline{x}} = \sum_{i=1}^{m} \underbrace{\lambda_i}_{\text{ALTEZZA}} \cdot \underbrace{(b_1^i - a_1^i)\cdot(b_2^i - a_2^i) \cdots (b_N^i - a_N^i)}_{\text{BASE}} =$$

DIFFERENZIALE

$$= \sum_{i=1}^{m} \lambda_i \int_{a_1}^{b_1} \int_{a_2}^{b_2} \cdots \int_{a_N}^{b_N} 1 \cdot \underbrace{dx_1\, dx_2 \ldots dx_N}_{\text{su } \mathbb{R}^N\text{, sono le COMPONENTI del diff.}}$$

Def: Sia $f : \mathbb{R}^N \to \mathbb{R}$ LIMITATA e a SUPPORTO COMPATTO, si dice INTEGRALE di RIEMANN <u>SE ESISTE</u> il NUMERO:

$$* = \int_{\mathbb{R}^N} f(\underline{x}) \cdot d\underline{x} = \int_{D_1} \int_{D_2} \cdots \int_{D_N} f(x_1, x_2, \ldots, x_N)\, dx_1\, dx_2\, dx_3 \cdots dx_N$$

il supporto di f è $K = D_1 \times D_2 \times \ldots \times D_N \subseteq \mathbb{R}^N$

$$① = \sup \left\{ \int_{\mathbb{R}^N} \varphi(\underline{x})\, d\underline{x} : \varphi \in S_-(f) \right\}$$

$$② = \inf \left\{ \int_{\mathbb{R}^N} \varphi(\underline{x})\, d\underline{x} : \varphi \in S_+(f) \right\}$$

e $① = ② = *$

PROPRIETÀ :

1) LINEARITÀ : Siano f e g INTEGRABILI su $\mathbb{R}^N$ allora

$$\int_{\mathbb{R}^N} (f+g)(\underline{x})\, d\underline{x} = \int_{\mathbb{R}^N} f(\underline{x})\, d\underline{x} + \int_{\mathbb{R}^N} g(\underline{x})\, d\underline{x}$$

Se $\lambda \in \mathbb{R}$: $\displaystyle\int_{\mathbb{R}^N} \lambda f(\underline{x})\, d\underline{x} = \lambda \int_{\mathbb{R}^N} f(\underline{x})\, d\underline{x}$

2) Se $f \leq g$ $(\forall \underline{x} \in \mathbb{R}^N \quad f(\underline{x}) \leq g(\underline{x}))$ e INTEGRABILI su $\mathbb{R}^N$

$$\int_{\mathbb{R}^N} f(\underline{x})\, d\underline{x} \leq \int_{\mathbb{R}^N} g(\underline{x})\, d\underline{x}$$

OSS : $f \leq |f|$ quindi $\displaystyle\int_{\mathbb{R}^N} f(\underline{x})\, dx \leq \int_{\mathbb{R}^N} |f(\underline{x})|\, d\underline{x}$

3) $\displaystyle\left| \int_{\mathbb{R}^N} f(\underline{x})\, d\underline{x} \right| \leq \int_{\mathbb{R}^N} |f(\underline{x})|\, d\underline{x}$

HW : $N=2$ $\quad f(x,y) = x^2 + y e^x$ definita in $K = \underbrace{[0,1]}_{x} \times \underbrace{[1,2]}_{y} \subseteq \mathbb{R}^2$

- $\displaystyle\int_K f(x,y)\, dx\, dy = ?$
- $\displaystyle\int_0^1 \int_1^2 f(x,y)\, dx\, dy \boxed{\overset{?}{=}} \int_1^2 \int_0^1 f(x,y)\, dx\, dy$

OPERAZIONE di CONVOLUZIONE $\mathbb{R}^N$

$N=1$ | $N=2$

Sia x SEGNALE IN INGRESSO

$x : \mathbb{R} \to \mathbb{R}$, $t \mapsto x(t)$ | $x : \mathbb{R}^2 \to \mathbb{R}$, $(u,v) \mapsto x(u,v)$

Sia w un NUCLEO o KERNEL

$w : \mathbb{R} \to \mathbb{R}$, $t \mapsto w(t)$ | $w : \mathbb{R}^2 \to \mathbb{R}$, $(u,v) \mapsto w(u,v)$

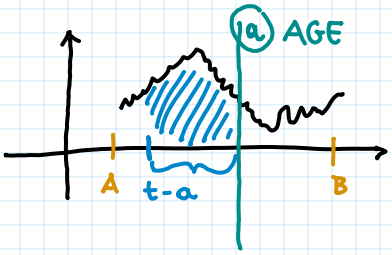
Si definisce l'OPERAZIONE di CONVOLUZIONE SCONTRANDO w con x:

$$(x * w)(t) = \int_{\mathbb{R}} x(\underbrace{a}_{\text{AGE}}) \cdot w(t-a)\, da = s(t) \quad \text{FEATURE MAP}$$

$$(x * w)(u,v) = \int_{\mathbb{R}^2} x(a,b) \cdot w(u-a, v-b)\, da\, db = s(u,v) \quad \text{FEATURE MAP}$$

OSS: Sia $x : \mathbb{R} \to \mathbb{R}$ un SEGNALE "RUMOROSO"

$\longrightarrow$ obiettivo: ottenere segnale pulito

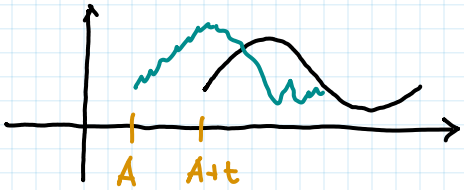


$$\int x(a) \cdot w(t-a)\, da$$

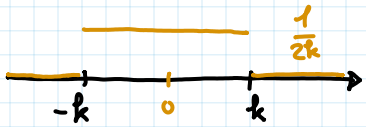
"SOMMA"

Dobbiamo chiedere che w sia una DENSITÀ di PROBABILITÀ

$$\begin{cases} w \geq 0 \\ \int_{\mathbb{R}} w(t)\, dt = 1 \end{cases}$$



FILTRO di MEDIA $N=1$



$$w(t) = \mathbb{1}_{[-k,k]}(t) \cdot \frac{1}{2k}$$

$x : \mathbb{R} \to \mathbb{R}$ generico

$$(x * w)(t) = \int_{\mathbb{R}} x(a)\, w(t-a)\, da = \int_{\mathbb{R}} x(\underbrace{a}_{t-\bar{t}}) \cdot \mathbb{1}_{[-k,k]}(\overbrace{t-a}^{\bar{t}=t-a}) \cdot \frac{1}{2k}\, da$$

$$= \int_{-k}^{k} x(t-\bar{t}) \cdot \frac{1}{2k}\, d\bar{t}$$

$a = t - \bar{t}$
$da = -d\bar{t}$

$$= \frac{1}{2k} \int_{-k}^{k} x(t-\bar{t})\, d\bar{t}$$

PROPRIETÀ :

1) COMMUTATIVA :

$$\boxed{(x * w)(t)} = \int_{\mathbb{R}} x(\underbrace{a}_{t-b})\, w(\underbrace{t-a}_{b})\, \underbrace{da}_{db=-da} = \int_{+\infty}^{-\infty} x(t-b)\, w(b)\, (-db)$$

$$= -1 \cdot \int_{+\infty}^{-\infty} x(t-b) \cdot w(b)\, db = \underbrace{\int_{-\infty}^{+\infty} w(b)\, x(t-b)\, db = \boxed{(w * x)(t)}}$$

2) LINEARITÀ

x, y segnali in input, $\lambda, \mu \in \mathbb{R}$, w kernel:

$$((\lambda x + \mu y) * w)(t) = \lambda (x * w)(t) + \mu (y * w)(t)$$

LINEARITÀ dell'INTEGRALE

3) La CONVOLUZIONE è INVARIANTE rispetto al TEMPO t

Considero un segnale $\tilde{x} : \mathbb{R} \to \mathbb{R}$ ritardato di τ rispetto a x

$$\tilde{x}(t) = x(t-\tau)$$

$$(\tilde{x} * w)(t) = \int_{\mathbb{R}} \tilde{x}(a)\, w(t-a)\, da = \int_{\mathbb{R}} x(\underbrace{a-\tau}_{b})\, w(\underbrace{t-a}_{t-b-\tau})\, da =$$

$b = a - \tau \quad \rightsquigarrow \quad \frac{db}{da} = 1$

$a = b + \tau$

$db = da$

$$= \int_{\mathbb{R}} x(b)\, w(\underbrace{\boxed{t - b - \tau}}_{\text{tempo}})\, db = s(t-\tau)$$

$$= (x * w)(t-\tau)$$

REGOLA del CAMBIO di VARIABILE di INTEGRAZIONE su $\mathbb{R}^N$

Siano $A, B \subseteq \mathbb{R}^N$, $\quad g : A \to B$, $\; \underline{x} \mapsto \underline{y}$ DIFFERENZIABILE, BIIETTIVA

Sia $f : \mathbb{R}^N \to \mathbb{R}$ INTEGRABILE in A, allora

$$\boxed{\int_A f(\underline{x})\, d\underline{x} = \int_B f(g(\underline{x}))\, |\det(J_g)|\, d\underline{y}}$$

g

$$g : \underset{\mathbb{R}^N}{A} \longrightarrow \underset{\mathbb{R}^N}{B}$$

$$\begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_N \end{bmatrix}$$

MATRICE JACOBIANA

$$J_g = \begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \cdots & \frac{\partial g_1}{\partial x_N} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & & \frac{\partial g_2}{\partial x_N} \\ \frac{\partial g_N}{\partial x_1} & & \cdots & \frac{\partial g_N}{\partial x_N} \end{pmatrix}$$

$N = 2$

$\alpha \overset{g_1}{=} u - a$

$\beta \underset{g_2}{=} v - b$

$$J_g = \begin{pmatrix} \frac{\partial g_1}{\partial a} = -1 & \frac{\partial g_1}{\partial b} = 0 \\ \frac{\partial g_2}{\partial a} = 0 & \frac{\partial g_2}{\partial b} = -1 \end{pmatrix}$$

PROPRIETÀ :

1) COMMUTATIVA

$$(x * w)(u,v) = \int_{\mathbb{R}^2} x(a,b) \cdot w(\overbrace{u-a}^{\alpha}, \overbrace{v-b}^{\beta})\, da\, db =$$

$$= \int_{\mathbb{R}^2} x(u-\alpha, v-\beta) \cdot w(\alpha,\beta) \cancel{|\det(J_g)|}^{1} \, d\alpha\, d\beta \qquad = (w * x)(u,v)$$

$$\det \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = 1$$

2) LINEARITÀ $\quad x, y$ segnali in input, $\lambda, \mu \in \mathbb{R}$, w kernel

$$((\lambda x + \mu y) * w)(u,v) = \lambda (x * w)(u,v) + \mu (y * w)(u,v)$$

3) La CONVOLUZIONE è INDIPENDENTE dalla POSIZIONE

$\tilde{x} : \mathbb{R}^2 \to \mathbb{R} \qquad \tilde{x}(u,v) = x(u-u_0, v-v_0)$

$$(\tilde{x} * w)(u,v) = \int_{\mathbb{R}^2} \underbrace{\tilde{x}(a,b)} \cdot w(u-a, v-b)\, da\, db$$

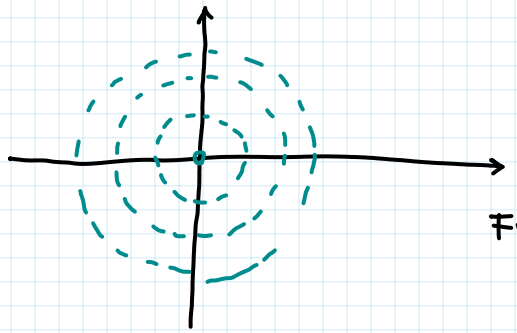
$$\int_{\mathbb{R}^2} x(\underbrace{a-u_0}_{\alpha}, \underbrace{b-v_0}_{\beta}) \cdot w(u-a, v-b)\, da\, db$$

$$g : \begin{cases} \alpha \overset{g_1}{=} a - u_0 \\ \beta \underset{g_2}{=} b - v_0 \end{cases} \qquad \det \begin{pmatrix} \frac{\partial g_1}{\partial a} = 1 & \frac{\partial g_1}{\partial b} = 0 \\ \frac{\partial g_2}{\partial a} = 0 & \frac{\partial g_2}{\partial b} = 1 \end{pmatrix} = 1$$

$$= \int_{\mathbb{R}^2} x(\alpha,\beta) \cdot w(\boxed{u - \alpha - u_0}, \boxed{v - \beta - v_0})\, d\alpha\, d\beta$$

$$= (x * w)(u - u_0, v - v_0)$$

FILTRO GAUSSIANO $N=2$: parametri μ, σ $\mathcal{N}(0,1)$



$$w(u,v) = \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{u^2+v^2}{2\sigma^2}}$$

Fisso : $\mu = 0$

$$= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{u^2}{2\sigma^2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{v^2}{2\sigma^2}}$$

$$(x * w)(u,v) = \int_{\mathbb{R}^2} x(a,b) \cdot w(u-a, v-b)\, da\, db =$$

$$= \int_{\mathbb{R}^2} x(u-a, v-b) \cdot \frac{1}{2\pi\sigma^2} e^{-\frac{\overbrace{a^2+b^2}^{\text{RAGGIO}^2}}{2\sigma^2}} da\, db$$

COMMUTATIVA

$$\boxed{\rho = \sqrt{a^2+b^2}}$$

$\theta \in [0, 2\pi)$

$$\begin{cases} a \overset{g_1}{=} \rho \cos\theta \\ b \underset{g_2}{=} \rho \operatorname{sen}\theta \end{cases}$$

HW.

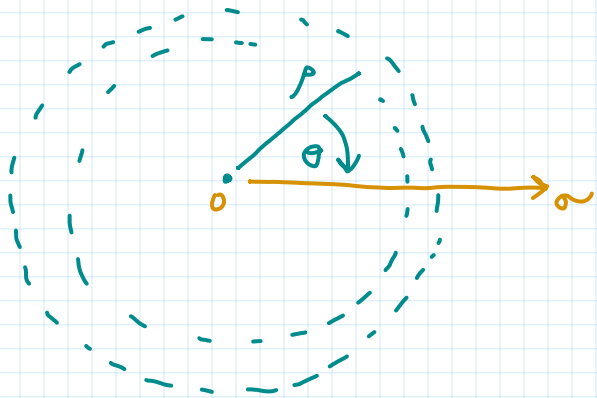
$$\frac{\partial g_1}{\partial a} \quad \frac{\partial g_1}{\partial b}$$

$$\frac{\partial g_2}{\partial a} \quad \frac{\partial g_2}{\partial b}$$

$$\det \begin{pmatrix} \cos\theta & -\rho \operatorname{sen}\theta \\ \operatorname{sen}\theta & \rho\cos\theta \end{pmatrix} = \rho\cos^2\theta + \rho \operatorname{sen}^2\theta = \rho(\cos^2\theta + \operatorname{sen}^2\theta) = \rho$$

$$= \boxed{\int_0^{+\infty} \int_0^{2\pi}} x(\underbrace{u-a}, \underbrace{v-b}) \boxed{\cdot \frac{1}{2\pi\sigma^2} e^{-\frac{\rho^2}{2\sigma^2}} \rho}\, d\rho\, d\theta$$

$u - \rho\cos\theta$, $v - \rho\operatorname{sen}\theta$



$\sim 3\sigma$ $\qquad$ piccolo a piacere

$$\exists\, M > 0 \quad \text{t.c.} \quad \rho > M \quad \Rightarrow \quad |w(u,v)| < \varepsilon$$

$$\int_{\mathbb{R}^2} \dots \approx \int_{-M}^{M}\int_{-M}^{M} \dots$$

$$(x * w)(u,v) \approx \int_{-M}^{M}\int_{-M}^{M} x(u-a, v-b) \cdot \frac{1}{2\pi\sigma^2} e^{-\frac{a^2+b^2}{2\sigma^2}} \, da\, db$$

$$= \int_{0}^{M}\int_{0}^{2\pi} \dots \, d\rho\, d\theta$$

↳ Nel DISCRETO, scelgo M e σ come parametri